

UMAN', Ukraine

WMO: 335870

Lat: 48.7669N

Lon: 30.2328E

Elev: 709

StdP: 14.32

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.7	(c) 5.7	(d) -5.7	(e) 4.2	(f) 0.1	(g) 0.0	(h) 5.6	(i) 7.0	(j) 28.4	(k) 26.1	(l) 27.1	(m) 25.7	(n) 2.4	(o) 320	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	88.9	68.7	85.8	67.8	82.9	66.4	71.1	83.8	69.7	81.7	68.3	79.3	3.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	102.8	76.4	65.6	97.0	75.0	64.2	92.2	73.5	35.4	84.2	34.1	81.7	32.9	79.3	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	18.4	12.2	-5.4	93.0	6.6	3.6	-10.1	95.6	-14.0	97.7	-17.6	99.8	-22.4	102.4
			-5.9	74.0	6.4	3.1	-10.6	76.2	-14.3	78.0	-17.9	79.7	-22.6	81.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	24.9	27.8	36.5	49.5	59.6	65.9	69.7	68.2	58.3	47.0	37.7	28.5
	DBStd	17.65	10.03	9.97	8.14	7.20	7.16	5.97	5.19	5.94	6.91	7.72	8.62	9.69
	HDD50	3127	777	621	423	93	10	0	0	0	12	149	376	666
	HDD65	6668	1242	1041	884	466	197	62	16	33	220	558	818	1131
	CDD50	2361	0	0	3	78	306	476	610	564	260	56	8	0
	CDD65	430	0	0	0	1	29	88	161	132	18	1	0	0
	CDH74	4394	0	0	0	24	362	848	1501	1415	237	7	0	0
	CDH80	1334	0	0	0	3	73	215	482	509	52	0	0	0
Wind	WSAvg	4.0	4.4	4.6	5.4	4.4	3.8	3.6	3.2	3.3	3.4	3.6	4.0	4.1
Precipitation	PrecAvg	24.8	1.8	1.5	1.5	1.8	2.1	3.5	3.3	2.3	2.0	1.5	1.8	1.7
	PrecMax	39.4	7.0	4.1	5.0	4.4	5.9	6.7	8.3	6.6	7.6	4.9	3.7	3.7
	PrecMin	16.9	0.3	0.1	0.1	0.4	0.1	0.4	0.6	0.2	0.0	0.1	0.4	0.3
	PrecStd	4.8	1.3	1.0	1.0	0.9	1.4	1.6	1.9	1.3	1.9	1.2	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	50.5	63.1	77.2	86.5	88.7	92.3	94.1	85.3	74.0	61.9	50.3
		MCWB	41.2	44.0	49.5	58.9	65.3	68.3	71.0	70.2	65.7	60.3	54.3	47.2
	2%	DB	41.8	45.3	57.0	71.2	81.6	85.3	88.5	89.7	80.4	68.1	56.0	46.7
		MCWB	39.1	41.1	46.4	55.2	63.2	67.4	69.3	68.7	63.5	58.0	51.7	44.6
	5%	DB	38.9	42.0	52.4	67.4	78.1	82.4	85.5	85.5	76.0	62.8	52.4	43.5
		MCWB	36.9	39.0	44.3	53.2	62.1	66.4	68.6	67.4	61.2	55.6	49.5	41.9
	10%	DB	36.5	39.1	48.5	63.3	74.2	79.1	82.1	81.9	71.5	58.6	49.6	40.3
		MCWB	35.0	36.8	42.2	51.2	60.2	65.4	67.3	65.8	59.7	53.3	47.3	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	45.5	50.4	60.7	67.4	72.1	73.5	73.1	68.3	62.7	55.5	48.3
		MCDB	44.0	48.6	60.8	76.5	79.9	82.5	85.8	88.4	79.5	71.4	59.2	49.5
	2%	WB	39.5	41.7	47.5	56.7	65.2	69.7	71.4	70.7	65.2	59.2	52.5	44.7
		MCDB	41.4	44.8	55.0	68.3	77.8	81.0	83.7	84.7	76.4	65.3	55.7	46.3
	5%	WB	37.1	39.3	45.1	54.6	63.5	68.2	70.1	68.9	62.8	56.5	50.0	42.0
		MCDB	38.5	41.5	51.2	64.5	74.8	79.0	81.9	81.2	73.3	62.3	52.1	43.4
	10%	WB	35.1	36.9	42.6	52.4	61.6	66.6	68.6	67.2	60.6	53.8	47.5	39.0
		MCDB	36.3	38.9	47.7	61.3	71.9	76.4	79.5	78.6	69.0	58.2	49.5	40.2
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.8	13.5	18.5	20.7	19.5	19.9	21.3	18.7	15.7	9.4	8.5
		MCDBR	8.5	12.5	21.3	25.8	26.2	24.5	24.9	27.3	27.0	23.2	13.6	10.2
		MCWBR	7.1	9.3	12.8	12.6	10.9	9.9	9.4	10.0	12.5	13.5	9.2	8.4
	5% WB	MCDBR	7.5	11.2	19.1	22.1	22.5	21.0	21.9	23.2	23.4	20.2	11.8	9.4
		MCWBR	6.6	9.0	12.3	11.7	10.3	9.8	9.3	9.4	12.5	12.6	8.7	8.1
Clear-Sky Solar Irradiance	taub		0.314	0.344	0.378	0.435	0.407	0.402	0.430	0.432	0.395	0.365	0.344	0.314
	taud		2.235	2.192	2.218	2.119	2.273	2.306	2.233	2.225	2.296	2.364	2.334	2.298
	Ebn at Noon		230	249	261	257	271	273	262	256	254	242	218	214
	Edh at Noon		24	32	37	46	41	40	42	41	34	27	23	21
All-Sky Solar Radiation	RadAvg		315	570	962	1416	1831	1952	1901	1697	1154	701	298	223
	RadStd		56	100	136	138	150	134	112	98	170	98	43	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.06	N/A	N/A	+1.78	N/A	N/A	N/A	N/A	+209	N/A
Regional (9 neighbors)	+1.24	N/A	N/A	+1.71	N/A	N/A	-232	-333	+200	+97

Nomenclature: See separate page